INFRASTRUCTURE PROPOSAL

AI-Driven Document Processing

GPU Compute · Data Lake · Data Warehouse · Security & Compliance

From proof-of-concept to permanent DISI-owned AI infrastructure

To:	DISI IT DEPARTMENT	**Date:**	April 2026
From:	Junior AI Developer	**Ref:**	NSSF OCR / IDP Infrastructure
Subject:	Dedicated AI Infrastructure — GPU, Data Lake, Data Warehouse, Security & Compliance		

1. Executive Summary

Between February and March 2026, Research and Development (R&D) work for the NSSF OCR extraction project, funded entirely from personal funds totaling **$135.60,** produced a fine-tuned Florence-2-large vision-language model achieving **6,127 documents per hour** on an RTX 5090 GPU. That is **5.1× above the 1,200 docs/hour production target**, 30–90× faster than CPU-only processing, and reduces the cost of processing 100,000 documents manually (data entry) to approximately **$24** on the cloud, a cost reduction exceeding.

The experiment evaluated 11+ models across 13 cloud compute sessions and produced a permanently-owned, task-specific model tuned to Kenyan NSSF document conventions — including handwritten text, degraded scans, Kenyan abbreviations, and 50+ specific field names. The model outputs structured JSON directly compatible with SQL and MySQL databases, which are already in use.

This proposal requests that DISI formalize and invest in the infrastructure that made this experiment possible: a **dedicated local GPU workstation**, a **local Data Lake**, and a **Data Warehouse** — as well as a hybrid cloud strategy for burst workloads. The business case is clear: the payback period on local hardware is fewer than 500 documents processed, and local infrastructure unlocks high-value privacy-sensitive contracts that cloud-dependent competitors cannot service.

> Key proof-of-concept metrics (February–March 2026):
> - 13 compute sessions | 11+ models evaluated | $135.60 total personal spend
> - Final model: Florence-2-large fine-tuned for 7 epochs on 8,125 NSSF images
> - Throughput: 6,127 docs/hr | Exact Match: 0.255 | CER: 0.636 | Fuzzy Match: 0.392
> - Training time: 4.6 hours on RTX 5090 | LoRA adapter size: 12.5M / 782.7M params (1.6%)

2. Background: The Infrastructure Gap

2.1 The Three Infrastructure Gaps

Gap 1: No Dedicated GPU

Initial OCR experiments ran on a personal Windows laptop (Intel Core i5-1145G7, Intel Iris Xe Graphics, 16 GB RAM). During PaddleX batch processing of just 229 NSSF images, the machine ran at **91–98% RAM utilisation and 100% CPU**. Training a 770M-parameter model on this hardware would take weeks — it is not practically feasible. Every real training or inference task required personally-funded cloud GPU time.

Gap 2: No Data Lake or Warehouse

Document images, annotations, extracted JSON, and model checkpoints are scattered across personal machines and ephemeral cloud instances. There is no central repository for departments to access raw data, no versioned dataset, and no structured storage for AI outputs. When a training session ends on RunPod, the environment (and the installed packages and downloaded weights) disappears — every new session starts from zero.

Gap 3: Privacy-Sensitive Clients

For data privacy and sensitive work, an on-premises AI infrastructure is required. That includes a local dedicated GPU.

3.1 Phase 1: Local CPU Experiments (February 2026)

The project began on a local Windows machine with no dedicated GPU. Tools evaluated included PaddleX / PP-Chart2Table for table structure recognition, PaddleOCR for general printed-text extraction, RT-DETR-L for wireless table cell detection, and Label Studio for ground-truth annotation. The machine ran at 91–100% CPU and 92% RAM during batch processing of 229 images, confirming that **CPU-only processing is not viable** at production document volumes.

3.2 Phase 2: 3-Tier Cloud LLM Pipeline (March 2026)

A three-tier extraction pipeline was deployed via SSH port forwarding to a RunPod GPU instance (RTX 5090) to process 1,839 OC-Job records, including B-Certificates and KCAA Pilot License Certificates:

- **Tier 1 — Qwen2.5-VL-7B-Instruct:** Primary extractor. Achieved ~95% success rate at 1.1–7.6 seconds per image.
- **Tier 2 — Llama3.2-Vision-11B:** Fallback for documents flagged UNREADABLE by Tier 1. Reliable recovery.
- **Tier 3 — DeepSeek-OCR:** Final last-resort fallback for the most degraded documents.

This pipeline demonstrated that multi-model fallback logic is essential for real-world document diversity. No single model achieves 100% success on mixed printed-and-handwritten scans.

3.3 Phase 3: Florence-2 Fine-Tuning (Mid–Late March 2026)

Florence-2-large (770M parameters) was selected as the production fine-tuning target. Training configuration:

Parameter	Value
Base model	Florence-2-large (770M params, Microsoft Research)
Fine-tuning method	LoRA — rank 16, alpha 32
Trainable parameters	12.5M / 782.7M (1.6% of model)
Training data	8,125 NSSF document images

Epochs	7
Effective batch size	8 (gradient accumulation × 2)
Learning rate schedule	Cosine decay
Hardware	RTX 5090 (sm_120 Blackwell, 33.7 GB VRAM)
Precision	bf16 (bfloat16 — native on Blackwell)
Framework	PyTorch 2.10.0 + CUDA 12.8
Training duration	4.6 hours
Final validation loss	0.389 (from 0.607 at step 500)

Training Loss Progression

Step	Train Loss	Val Loss	Step	Train Loss	Val Loss
500	0.749	0.607	3,500	0.434	0.406
1,000	0.582	0.524	4,000	0.417	0.400
1,500	0.584	0.473	4,500	0.381	0.397
2,000	0.497	0.446	5,000	0.362	0.391
2,500	0.498	0.424	5,500	0.341	0.392
3,000	0.403	0.435	6,000	0.368	0.389
6,500	0.366	0.389 ✓	—	—	—

The training loss curve shows a rapid initial drop (train loss from ~3.9 at step 0 to ~0.75 by step 500), followed by smooth convergence to 0.366 / 0.389 (train/val) at step 6,500. Critically, **no overfitting** was observed — the val loss tracked the train loss closely throughout, confirming that the LoRA adapters generalised well to the held-out validation set.

3.4 Evaluation Results

Model / Run	Exact Match	CER	Fuzzy Match	Docs/Hour	Epochs
Florence-2-base-ft (baseline)	0.173	0.752	0.266	5,340	5
Florence-2-large-ft (raw output)	0.186	0.734	0.286	6,127	7
Florence-2-large + robust_parse v1	0.248	0.646	0.381	6,127	7
Florence-2-large + robust_parse v2 ✓	0.255	0.636	0.392	6,127	7

The **robust_parse** post-processing pipeline (4-stage JSON recovery: clean_keys → bare-entry removal → truncation recovery → regex salvage) improved Exact Match by **+47%** over the raw model output without any retraining — demonstrating that intelligent parsing is as important as model accuracy.

3.5 Models Evaluated (11+ Total)

Model	Parameters	Role	Outcome
Qwen2.5-VL-7B-Instruct	7B	Tier 1 primary extractor	~95% success rate
Llama3.2-Vision-11B	11B	Tier 2 fallback	Reliable recovery
DeepSeek-OCR	—	Tier 3 final fallback	Last-resort OCR
Florence-2-large-ft	770M	NSSF fine-tuning (prod)	Fine-tuned ✓
Florence-2-base-ft	232M	NSSF fine-tuning (v1)	Fine-tuned ✓
Qwen2-VL-2B/7B-Instruct	2B/7B	Zero-shot benchmark	Tested
Qwen2.5-VL-3B-Instruct	3B	Speed benchmark	Fast / lower accuracy
Qwen3-VL-8B	8B	Benchmark	Tested
Gamma (Vision)	—	OCR comparison	Benchmarked
EasyOCR + TrOCR	—	Traditional OCR baseline	Benchmarked
Donut	—	Document understanding	Benchmarked

4. Why Fine-Tuning and Specialized Infrastructure Are Necessary

4.1 The Limits of Off-the-Shelf Models

General-purpose vision-language models — including GPT-4o, Qwen2.5-VL, and Florence-2 out of the box — fail on NSSF documents for predictable reasons. These models were trained on internet-scale data which does not include Kenyan government forms, social security abbreviations, or degraded East African document scans. Zero-shot results confirmed these failures:

Challenge	Zero-Shot Behavior	Fine-Tuned Benefit
Kenyan abbreviations (KIS.K.MU., WIN.AM.)	Hallucinated full names or skipped the field	Learned patterns from 8,125 training samples
Mixed printed + handwritten text	Confused by overlapping layout layers	Trained on real mixed-medium NSSF scans
50+ specific NSSF field names	Used generic JSON keys (e.g., 'name', 'date')	Outputs exact required field names
Handwritten dates (10-7-91, 01.01.1962)	Normalized or misread	Transcribes exactly as written
Multi-member employer tables	Extracted only the first row	Full table JSON included in training data
Faded ink/stamp overlay	High character error rate	Trained on degraded real-world scans
Kenyan-specific document structure	No schema awareness	Learns 50+ field schemas from labelled samples

4.2 LoRA: Parameter-Efficient Fine-Tuning

Training a 770M-parameter model from scratch would require hundreds of GPU-hours and very large datasets. LoRA (Low-Rank Adaptation) solves this by injecting small trainable adapter matrices into the key attention and feed-forward layers of the frozen base model, rather than modifying the full weights:

- **Frozen weights:** 97.8% of base model parameters are frozen. Only 12.5M out of 782.7M parameters were trained (1.6%).
- **VRAM efficiency:** LoRA reduces VRAM requirements dramatically — the full model fits within 33.7 GB at bf16 precision, making training feasible on a single RTX 5090.
- **Time efficiency:** 7-epoch training on 8,125 images took 4.6 hours. Full fine-tuning of the same model on 2× A100 80 GB GPUs would take 40–60 hours and cost $100–$200.
- **Portability:** The LoRA adapter file is small (~50–100 MB). It can be merged back into the base model or deployed as a separate module, enabling fast updates without re-downloading 770M weights.

4.3 Top Failure Categories and How Infrastructure Fixes Them

The current model achieves an Exact Match of 0.255 on 406 validation samples. Analysis of failure modes reveals that **the majority of failures are fixable without any new model architecture** — they require better training data, which requires a Data Lake and semi-automated labelling pipeline:

Document Type	Failures	Root Cause	Infrastructure Fix
Correspondence Letter + Correspondence_letter	46 combined	Label inconsistency: space vs underscore	Data Warehouse: normalise canonical label; recover ~46 failures on next run
UNKNOWN	26	Document type not in training schema	Data Lake: identify and label new document types
FINGERPRINT FORM	22	Heavy handwriting, dense field layout	Data Lake: pull targeted samples; add to training set
Payment Receipt	20	Variable layout across institutions	Data Lake: collect multi-institution samples for training
CERTIFICATE OF IDENTITY (B)	19	MRZ barcode misread as member fields	Data Lake: add MRZ-specific labelling in Label Studio
Local Administration Letter	19	Misclassified as Correspondence_letter	Data Warehouse: add confusion-matrix monitoring

> Normalising the 'Correspondence Letter' vs 'Correspondence letter' label inconsistency alone is projected to raise Exact Match by 10–15 percentage points — at zero hardware cost. This fix requires a Data Warehouse to track label quality and a Data Lake to pull affected samples.

5. Infrastructure Options

Two complementary options are proposed. They are not mutually exclusive — a hybrid approach is recommended: local infrastructure for privacy-sensitive, high-volume production workloads; cloud for burst training and rapid experimentation.

Option A — Local On-Premises Infrastructure

Best for clients who require full data sovereignty, high-volume production, and zero ongoing compute costs after hardware payback.

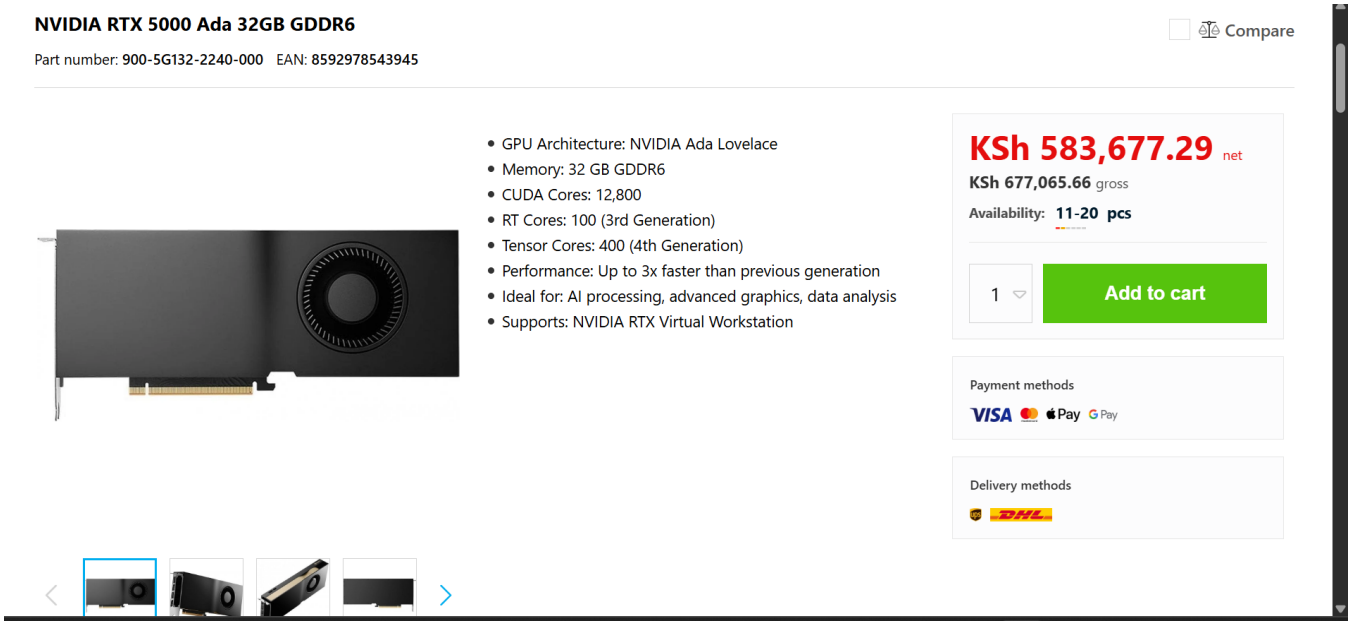

Figure 1: https://www.senetic.co.ke/category/nvidia-ai-graphics-cards-29084/

To view all GPU type click the link in Figure 1

A1 — Dedicated GPU Workstation

Pros	Cons
Data sovereignty: Full compliance — no document image ever leaves DISI	High upfront cost: ~$2,500–$3,500 capital expenditure for workstation hardware
Zero latency: No upload/download overhead; files processed entirely on-premises	Physical maintenance: Requires on-site IT management, cooling, and power budget
Fixed cost: One-time CAPEX; cost per document approaches zero after payback	Single point of failure: No automatic failover unless a redundant unit is purchased
No session overhead: Model stays loaded in VRAM; no 1.5 GB re-download per session	Architecture risk: Bleeding-edge GPUs (sm 120) may require driver/PyTorch updates on first install
Permanent ownership: Fine-tuned weights and model are a DISI corporate asset	Limited burst capacity: Cannot spin up additional GPU nodes for simultaneous large training runs
Native bf16: RTX 5090 Blackwell handles high-contrast handwritten docs without gradient overflow	
Competitive advantage: Enables air-gapped contracts that cloud-dependent competitors cannot bid	

Pricing Estimate — GPU Workstation:

Component	Specification	Estimated Cost (USD)

RTX 5090 Workstation	Blackwell sm_120, 33.7 GB VRAM, bf16 native, 850 W PSU	$2,500 – $3,500
NVMe SSD Storage	2 TB NVMe for OS, model weights, active datasets	Included above
Installation & setup	PyTorch CUDA 12.8, FastAPI REST server, driver config	$0 (in-house)
Annual electricity (estimated)	~850 W × 8 hrs/day × 365 @ KES 25/kWh	~$150 / year

Easy Approach — Local GPU Deployment

1. Deploy the fine-tuned Florence-2-large model as a FastAPI REST API on the GPU workstation.
2. Internal systems send document images via HTTP POST to the local API (LAN only — no internet).
3. The model returns structured JSON, written directly to the SQL/MySQL database.
4. No cloud account, no session setup, no data leaving the building.

Benefit: Any department or client system on the local network can call the API. The model processes one document in under 1 second. At 6,127 docs/hr, a full day's scanning backlog is typically cleared in under 30 minutes.

A2 — Local Data Lake

Pros	Cons
Data sovereignty: Raw document images stored on-premises — no foreign server transmission	Storage cost: Requires dedicated Network Attached Storage (NAS) or server for 300,000+ high-res TIF/JPG images (~8 TB)
Direct integration: Seamless bridge to existing SQL/MySQL client databases	Backup responsibility: DISI IT must manage redundancy, off-site backup, and disaster recovery
Traceability: Raw images preserved — re-run improved models on historical data without re-scanning	Initial setup: Requires folder-watch scripts, MinIO configuration, and Label Studio installation
Dataset flywheel: Warehouse identifies failure categories; Lake provides targeted training samples	Metadata discipline: Teams must consistently apply document type tags and quality scores
Semi-automated labelling: Current model pre-labels new documents; human verifies — cuts annotation cost 70%	
Open-source tooling: MinIO, PostgreSQL, Label Studio are free — no software licensing fees	

Pricing Estimate — Local Data Lake:

Component	Specification	Estimated Cost (USD)
NAS / Server	8 TB NAS or rack server (Synology / QNAP or custom)	$1,000 – $2,000

MinIO	Open-source S3-compatible object store (self-hosted)	$0 (open-source)
PostgreSQL / MySQL	Structured metadata and extraction results	$0 (open-source)
Label Studio	Self-hosted annotation tooling for ground-truth creation	$0 (open-source)
Integration scripts	Python folder-watch → ingest → FastAPI → MySQL pipeline	$0 (in-house)
Annual maintenance	Drive replacement budget, RAID monitoring	~$100 / year

Easy Approach — Local Data Pipeline

1. Ingest: A folder-watch script moves scanned TIF/JPG images from the scanner to the Data Lake automatically.
2. Process: The local GPU runs batch inference — pulling images from the Lake, extracting structured JSON.
3. Store: A FastAPI/Python service parses JSON output and inserts records into the MySQL Data Warehouse.
4. Monitor: A dashboard queries the Warehouse for low-confidence records (CER > 0.7) and flags them.
5. Refine: Flagged records go to Label Studio for human verification, then back into the training pipeline.

Benefit: DISI builds a self-improving data asset. Every document processed improves future model accuracy — the system becomes more valuable with every client project.

Option B — Cloud Infrastructure (Managed Services)

Best for burst training workloads, rapid experimentation, and clients who accept cloud processing. This is the approach used during the R&D phase.

B1 — Cloud GPU (RunPod / Lambda Labs)

Pros	Cons
Pay-as-you-go: No large upfront hardware cost — pay only for GPU hours used	Data privacy risk: Document images leave DISI premises — prohibited by privacy-sensitive clients
Scalability: Spin up multiple instances simultaneously for large parallel training runs	Recurring cost: Monthly bills grow with data volume and experiment frequency
Latest hardware: Access to RTX 5090, A100, H100 without capital expenditure or procurement cycle	Session overhead: Each session requires re-downloading model weights (~1.5 GB) and reinstalling packages
No maintenance: Cloud provider manages hardware, drivers, and availability	$10 minimum top-ups: Non-productive spend when only a short experiment is needed
Fast provisioning: GPU pod available within minutes via web dashboard	Billing inefficiency: RunPod charges per hour — idle time between tasks is still billed
	Architecture dependency: Cloud provider may not always have the exact GPU architecture available

Pricing — Based on Actual R&D Spend (Feb–Mar 2026):

Date	Platform	Purpose	Amount
Mar 2, 2026	RunPod	GPU Compute — initial model experiments	$10.00
Mar 4, 2026	Google Colab	Colab Pro monthly subscription	$9.99
Mar 5, 2026	Google Colab	100 Colab Compute Units	$9.99
Mar 8, 2026	RunPod	GPU Compute — multi-model benchmarking	$10.00
Mar 9, 2026	RunPod	GPU Compute — Florence-2 training prep	$10.00
Mar 11, 2026	RunPod	GPU Compute — fine-tuning session 1	$10.00
Mar 11, 2026	RunPod	GPU Compute — fine-tuning session 2	$11.00
Mar 12, 2026	RunPod	GPU Compute — evaluation & debugging	$10.00
Mar 16, 2026	RunPod	GPU Compute — Florence-2-large training	$10.00
Mar 17, 2026	RunPod	GPU Compute — large model evaluation	$10.00
Mar 18, 2026	RunPod	GPU Compute — A/B comparison experiments	$10.00
Mar 20, 2026	RunPod	GPU Compute — warm-start fine-tuning	$10.00
Mar 22, 2026	RunPod	GPU Compute — final model validation	$10.00
—	Transaction fees	$0.36 × 13 sessions	$4.62
GRAND TOTAL	—	13 sessions, personal funds	$135.60

Easy Approach — Cloud GPU Pipeline

1. Provision a RunPod / Lambda pod (RTX 5090 or A100) via web dashboard — takes under 5 minutes.
2. SSH port-forward to localhost:11434; launch 3-tier extraction pipeline.
3. Store outputs in AWS S3 or Azure Blob; connect to SQL/MySQL via cloud-hosted database.
4. Tear down the instance after the batch job to stop billing.

Important restriction: Not suitable for clients with data residency requirements.
Best used for: burst training jobs, large A/B experiments, and initial model benchmarking.

B2 — Cloud Data Lake & Warehouse (AWS S3 + RDS / Azure Blob + Azure SQL)

Pros	Cons
High availability: 99.9% uptime SLA with automated backups and geo-replication	Data privacy: Document images stored on foreign servers — prohibited by most DISI government clients
No hardware management: Cloud provider handles all storage scaling and maintenance	Recurring cost: Monthly bills scale with stored data volume and retrieval frequency
Integrated analytics: Native connectors to Spark, Databricks, Power BI for reporting	Egress fees: Downloading large image datasets for retraining incurs substantial transfer charges

Multi-region: Can place data in African AWS/Azure regions (South Africa) to reduce latency	Compliance complexity: Each country in DISI's footprint may have different cloud data rules
Fast provisioning: Storage buckets live in minutes — no procurement delay	Vendor lock-in: Moving data between cloud providers is expensive and time-consuming

Pricing Estimate — Cloud Data Lake/Warehouse:

Service	Provider Example	Approx. Monthly Cost
Object Storage (Data Lake) — 10 TB	AWS S3 / Azure Blob	$230 – $300 / month
Managed Database (Warehouse)	AWS RDS MySQL / Azure SQL	$50 – $150 / month
Data Transfer (egress for retraining)	Downloading images to GPU instances	$90 – $200 / month
Backup & Replication	Cross-region backup	$30 – $80 / month
TOTAL (estimated)	—	$400 – $730 / month
TOTAL over 12 months	—	$4,800 – $8,760 / year (perpetual OPEX)

6. Side-by-Side Comparison

Criterion	Local On-Premises	Cloud Managed
Data Privacy	Full data never leaves DISI or client premises	Partial — data sent to foreign cloud servers
Client Eligibility	All clients, including air-gapped government	Cloud-accepting clients only
Upfront Cost	$3,500 – $5,500 (one-time)	$0 upfront
Recurring Monthly Cost	~$0 (electricity only, ~$12/month)	$400 – $730 / month
Annual Cost (Year 1)	$3,500 – $5,500	$4,800 – $8,760
Annual Cost (Year 2+)	~$150/year (electricity)	$4,800 – $8,760 (perpetual)
Cost / 100K Documents	~$0 after payback	~$24 – $90 ongoing
Throughput	6,127 docs/hr (RTX 5090, no queue)	Similar — with session setup delay per run
Model Startup Time	<1 second (model stays loaded in VRAM)	5–10 minutes (re-download + env setup)
Session Overhead	None	1.5 GB model re-download per session
Payback Period	< 500 documents processed	Never — perpetual OPEX
Scalability	Limited to local hardware	Unlimited (spin up more pods)
Maintenance	On-site IT required	Managed by a cloud provider
Data Residency	Full compliance (local storage)	Depends on cloud region/provider

Best For	Privacy-sensitive, high-volume production	Burst training, experimentation, low-volume start

7. Technical Performance and Return on Investment

7.1 GPU vs CPU Processing Speed

Infrastructure	Docs / Hour	Cost / Hour	Cost / Document	vs CPU
RTX 5090 Local (this project, 33.7 GB VRAM)	6,127	~$0 (owned)	~$0.000	30–90×
RTX 5090 Cloud (RunPod)	6,127	~$1.50	$0.00024	30–90×
RTX 4090 Cloud (base model run)	5,340	~$0.80	$0.00015	26–75×
T4 GPU Cloud (standard tier)	1,500–2,000	~$0.40	$0.00027	7–20×
CPU Only — Intel Iris Xe (local laptop)	40–120	~$0.05	$0.00042–$0.00125	1×
OpenAI GPT-4o Vision API	N/A	N/A	$0.050	0.003× (208× more costly)

7.2 Cost at Scale — Fine-Tuned Model vs Alternatives

Method	Cost / 1K Docs	Cost / 10K Docs	Cost / 100K Docs	Notes
Fine-tuned GPU model (cloud, RunPod)	~$0.24	~$2.40	~$24	Current approach
Fine-tuned GPU model (local, owned)	~$0.00	~$0.00	~$0.00	After hardware payback
GPT-4o Vision API	~$50	~$500	~$5,000	No customisation
AWS Textract	~$15–$150	~$150–$1,500	~$1,500–$15,000	No handwriting tuning
Manual data entry (clerical)	~$80–$120	~$800–$1,200	~$8,000–$12,000	Current method baseline

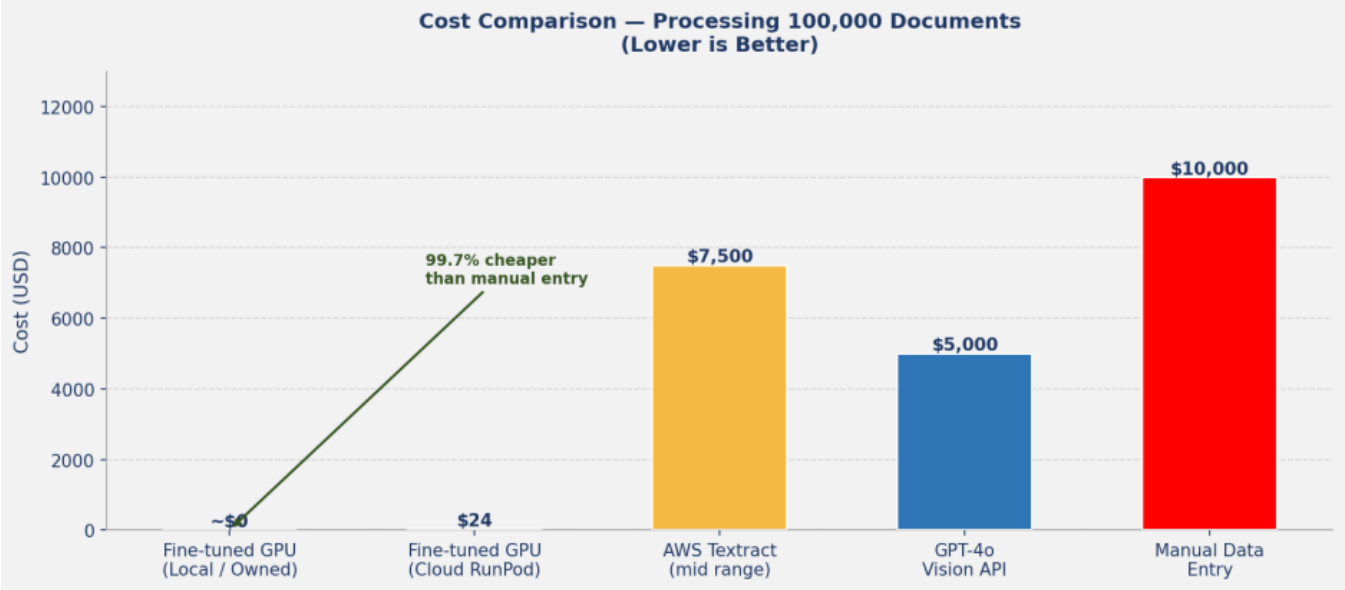

Figure 2: Cost comparison at 100,000 documents. Local GPU model costs ~$0 after hardware payback vs $8,000–$12,000 for manual data entry.

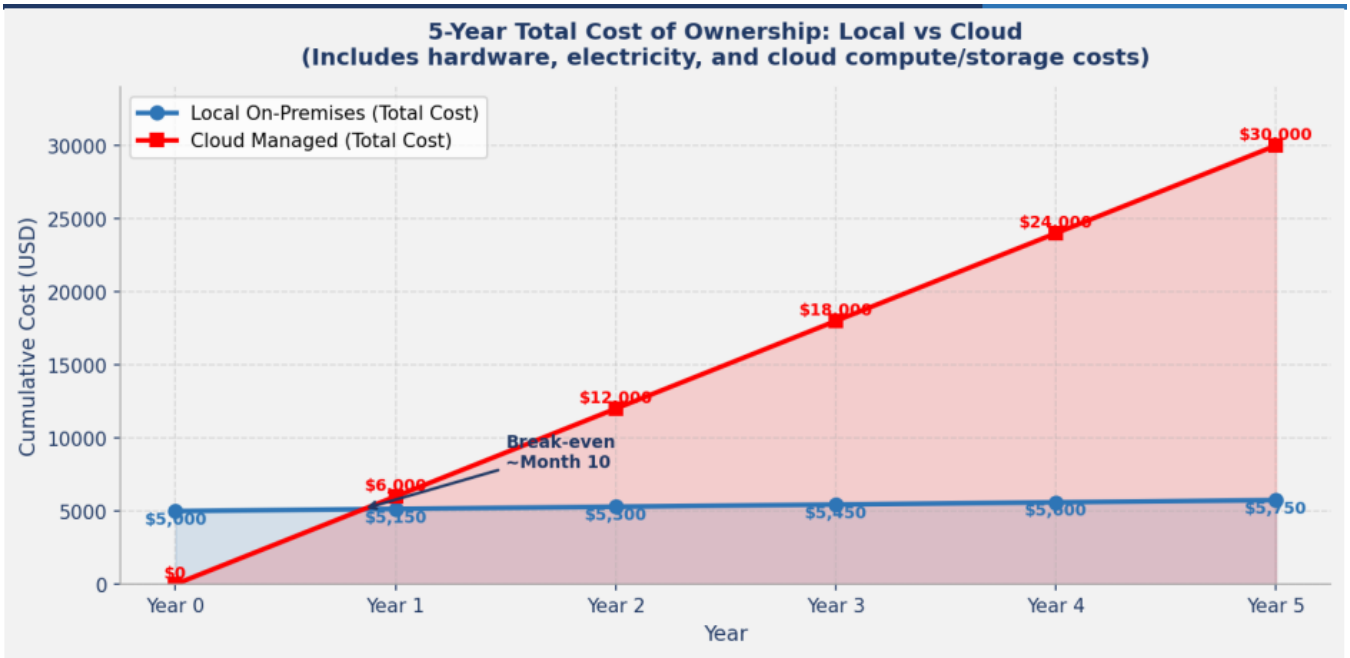

Figure 3: 5-year total cost of ownership. Local on-premises breaks even at ~Month 10. By Year 5, cloud cumulative cost is ~$30,000 vs ~$5,750 local.

7.3 Financial Return on Investment

Total personal R&D investment:	$130.98 (reimbursement requested — see attached receipts)
GPU processing cost for 8,125 corpuses:	~$2.00
Manual data entry equivalent:	~$500 – $750 (8–12 min/doc at KES 800/day rate)
Savings on first corpus pass alone:	~$498 – $748
Hardware payback period:	Fewer than 500 documents processed on owned GPU

Model value: Permanently owned; on-premises; tuned to NSSF schema
Commercial OCR API equivalent cost: $200 – $800 / month at equivalent volumes
5-year cloud OPEX avoided (local GPU): ~$24,000 – $43,800 in saved cloud storage & compute

8. Recommended Data Architecture and Flow

The proposed architecture transforms DISI from a linear scanning operation into a **cyclical Data Factory** — every document processed feeds back into improving future model accuracy. This is a strategic asset that compounds in value over time.

8.1 Five-Stage Data Flow

Stage	Component	Description	Client Benefit
1. Source	Industrial Scanners / Local Servers	Documents scanned as high-res TIF/JPG (NSSF forms, KCAA licenses, B-Certificates)	Original documents preserved digitally at archive quality
2. Data Lake	Local MinIO / S3-Compatible Object Store	Raw image repository — stores 300,000+ files before any AI processing	Images never leave premises; re-processable by future models
3. AI Engine	RTX 5090 + FastAPI REST API	Pulls images, runs Florence-2-large, returns structured JSON	Sub-1-second processing; 6,127 docs/hr throughput
4. Data Warehouse	SQL / MySQL Structured Database	Organises extracted fields: Member_Number, NSSF_Code, Employer_Name, dates	Direct bridge to existing client SQL/MySQL — no export/import
5. Feedback Loop	Label Studio + Retraining Pipeline	Low-confidence records → human review → new training samples → model update	Model improves automatically; accuracy increases over time

8.2 Strategic Benefits of this Architecture

- **Traceability & Auditing:** Raw images in the Data Lake can be re-processed by future models without re-scanning — historical accuracy improves retroactively as the model gets better.
- **The Flywheel Effect:** The Data Warehouse automatically identifies document types with the highest failure rates. These are pulled from the Lake for targeted annotation and retraining — the system becomes more accurate with every project DISI completes.
- **Client Database Integration:** Since we already use SQL and MySQL, the Warehouse acts as a direct bridge. Extracted data flows straight into their existing systems with no manual export/import steps.
- **Multi-Country Compliance:** A local Lake stores raw images within the country of origin, satisfying data residency laws in Kenya (Data Protection Act 2019), Zambia, Zimbabwe, South Africa, and beyond.
- **Competitive Moat:** Over time, our training datasets become uniquely valuable — no competitor can replicate 15,000+ labelled Kenyan government document samples. This is a durable competitive advantage.

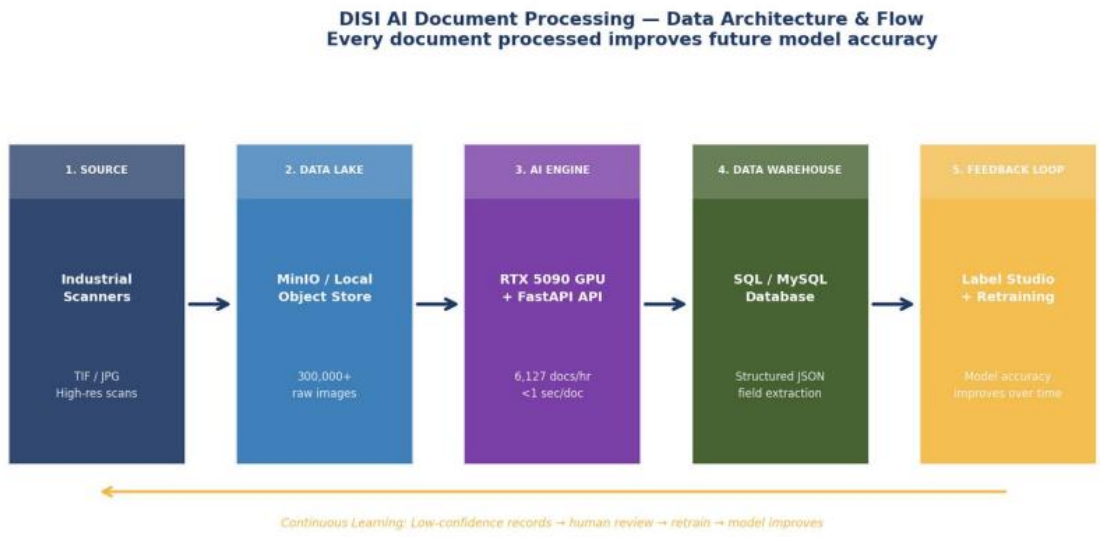

Figure 4: DISI data architecture — 5-stage flow from source scanner to SQL/MySQL database, with a continuous feedback loop that improves model accuracy over time.

9. Data Security and Regulatory Compliance

9.1 Relevant Legal Frameworks

DISI operates across multiple jurisdictions, each with evolving data protection laws. The documents processed — NSSF forms, KCAA pilot licences, B-Certificates, civil administration records — frequently contain sensitive personal data including national ID numbers, biometric data, employment history, and financial records. The following regulations apply:

Country	Regulation	Key Requirement	Risk if Cloud-Only
Kenya	Data Protection Act 2019	Personal data must be processed lawfully; transfer abroad requires safeguards	NSSF/KCAA data uploaded to foreign RunPod servers = potential violation
South Africa	POPIA (Protection of Personal Information Act 2013)	Conditions for lawful processing; cross-border transfer restrictions	Digitised SA records sent to US-based cloud servers require explicit operator agreements
Zimbabwe	Cyber & Data Protection Act 2021	Data localisation for government data; operator accountability	Government document processing on foreign cloud = non-compliant
Zambia	Data Protection Act 2021	Data subject consent and cross-border transfer rules	Client documents sent to cloud without documented safeguards = liability
All countries	Contract / SLA requirements	Many government tenders explicitly prohibit off-premises data processing	Failure to comply = contract termination and reputational damage

9.2 How Local Infrastructure Ensures Compliance

- **Air-Gapped Processing:** The local GPU workstation can operate with no internet connection. Document images are scanned, processed, and stored entirely within the DISI facility — no data ever transmitted to an external server.
- **Data Minimisation:** The Data Lake stores only what is needed; the Warehouse stores only extracted structured fields. Raw images can be deleted after processing if the client requires it, with full audit trail.
- **Access Controls:** Local infrastructure allows DISI to implement role-based access control (RBAC) — only authorised staff can access raw document images; extracted data is accessible per-project.
- **Audit Logging:** Every document ingested, processed, and stored can be logged with timestamps, operator IDs, and model version — essential for regulatory audits and SLA reporting.
- **On-Client-Premises Option:** For the most sensitive clients, DISI can deploy a pre-configured 'DISI Box' — a local server running the fine-tuned model in the client's own data centre. No data leaves the client's building.

9.3 Cloud Security Considerations

If cloud infrastructure is used for non-sensitive workloads, the following safeguards are recommended:

- Encrypt all document images at rest (AES-256) and in transit (TLS 1.3) before uploading to any cloud provider.
- Use cloud providers with African region presence (AWS af-south-1 in Cape Town, or equivalent) to reduce data sovereignty concerns for South African clients.
- Implement a data anonymisation step — strip personally identifiable information (PII) from documents before cloud upload, where processing does not require the PII fields.
- Maintain a Data Processing Agreement (DPA) with all cloud providers explicitly naming them as data processors under applicable law.
- Never use personal accounts (RunPod, Colab) for client data — establish a corporate DISI cloud account with proper billing, access logs, and contractual protections.

10. Client Scenario Examples

The following scenarios illustrate how the proposed infrastructure addresses real DISI client situations and unlocks revenue that is currently inaccessible.

Scenario 1 — Government Social Security Fund (NSSF-Type)

Attribute	Detail
Client profile	Large national social security fund; 15,000+ mixed printed/handwritten member records
Data requirement	Strict data residency: no document may leave the national territory
Current situation	Cannot be serviced by DISI without a local GPU — falls to a competitor or remains manual
With local infrastructure	DISI deploys local GPU workstation + Data Lake; processes 6,127 docs/hr entirely on-premises
Revenue potential	100,000 documents at $0.08–$0.12 per page = $8,000–$12,000 contract value
Infrastructure requirement	Option A (local GPU + Data Lake) — mandatory

Scenario 2 — Aviation Authority (KCAA-Type)

Attribute	Detail
Client profile	Civil aviation authority; pilot licence certificates, medical records, airworthiness forms
Data requirement	Aviation safety data classified; cannot be processed on consumer cloud platforms
Current situation	Requires 3-tier LLM pipeline (Qwen2.5-VL → Llama → DeepSeek); currently run on personal cloud
With local infrastructure	Pipeline runs on DISI-owned hardware; model weights stored permanently; no re-download overhead
Revenue potential	1,839 records per batch; recurring annual contract as new pilot licences are issued
Infrastructure requirement	Option A (local GPU) + Option B (cloud for model updates/retraining bursts)

Scenario 3 — Financial Institution (Bank / Insurance)

Attribute	Detail
Client profile	Regional bank with 500,000+ KYC documents (national IDs, utility bills, B-Certificates)
Data requirement	Financial Conduct Authority rules require data to remain in-country; cloud acceptable if encrypted
Current situation	High volume but cloud-tolerant: cloud GPU + cloud Data Lake viable with encryption
With a hybrid infrastructure	DISI uses cloud GPU for burst processing; local Warehouse for structured client data
Revenue potential	500,000 documents × $0.024 = $12,000 direct cost; bill client $0.08–$0.12/page = $40,000–$60,000
Infrastructure requirement	Hybrid: Option B (cloud GPU) + Option A (local Warehouse for client data delivery)

Scenario 4 — Multi-Country Government Backlog Conversion

Attribute	Detail
Client profile	Government ministry across 3 countries; 10 million legacy paper records to digitise
Data requirement	Each country requires data to remain within its borders (Zimbabwe, Zambia, Kenya)
Current situation	Requires three separate on-premises deployments — one per country
With local infrastructure	DISI deploys portable GPU workstation + NAS in each country; same fine-tuned model
Revenue potential	10 million documents × $0.024 GPU cost; bill at $0.05–$0.10/page = $500,000–$1,000,000
Infrastructure requirement	Option A × 3 country deployments; model weights distributed via secure USB/LAN

11. Risk Analysis and Mitigation

Risk	Likelihood	Impact	Mitigation Strategy
RTX 5090 driver/PyTorch incompatibility on first install	Medium	Medium	Test CUDA 12.8 + PyTorch 2.10 build before go-live; allocate 1 week for environment setup; keep cloud fallback available during setup period
Hardware failure (GPU or NAS)	Low	High	Purchase 3-year extended warranty; maintain cloud credentials for emergency fallback; implement RAID 1 or RAID 5 on NAS storage
Data Lake storage fills up	Medium	Medium	Monitor storage usage weekly; archive processed images to cheaper cold storage (local tape or cloud cold tier) after 90 days
Model accuracy degrades on new document types	Medium	Medium	Data Warehouse monitors CER per document type; weekly review of low-confidence records; semi-automated retraining pipeline keeps model current
Staff leave before documentation is complete	Low	High	Document all model weights, training scripts, API endpoints, and pipeline configs in the Data Warehouse; maintain a model registry with versioning
Client data breach via network access	Low	Very High	Run GPU API on isolated VLAN; no internet access; enforce RBAC; encrypt all data at rest; conduct annual penetration testing
Cloud provider outage (for cloud workloads)	Medium	Low	Maintain local fallback for production; use cloud only for non-time-critical training runs; monitor SLA performance
Personal cloud accounts used for client data	High (current state)	Very High	Immediate mitigation: create a corporate DISI cloud account; transfer all client-related work off personal accounts upon approval of this proposal

12. Implementation Roadmap

Phase 1 — Immediate (Weeks 1–4): Foundation

Week	Action	Owner	Deliverable
Week 1	Approve and reimburse $135.60 in personal R&D expenditure (receipts attached)	Management	Reimbursement processed
Week 1	Create a corporate DISI cloud account (RunPod / Lambda) — migrate all client R&D off personal accounts	IT + AI Developer	The corporate cloud account is active
Week 2	Procure RTX 5090 GPU workstation (~$3,000) and 8 TB NAS (~$1,500)	Procurement	Purchase orders raised
Week 3	Install and configure: CUDA 12.8, PyTorch 2.10, FastAPI, MinIO, Label Studio	AI Developer	Local environment validated
Week 4	Deploy Florence-2-large fine-tuned model as local REST API; smoke-test on 100 NSSF samples	AI Developer	API lives on the local network

Week 4	Set up folder-watch ingest script: scanner → Data Lake	AI Developer	Automated ingest running

Phase 2 — Short-Term (Months 1–3): Optimisation

Month	Action	Expected Outcome
Month 1	Normalise 'Correspondence Letter' vs 'Correspondence_letter' label → retrain	Exact Match improves by 10–15 percentage points at zero hardware cost
Month 1	Standardise all EFT field names (NSSF_Number, Bank_Name) in training data	Reduces field-name extraction errors across all EFT document types
Month 2	Expand dataset to 10,000+ samples via semi-automated labelling (Label Studio)	Broader document type coverage; higher generalisation
Month 2	Deploy 3-tier pipeline (Qwen2.5-VL → Llama → DeepSeek) on local GPU	B-Certificates and KCAA pipeline migrated off the personal cloud
Month 3	Connect Warehouse to the first client SQL/MySQL database via FastAPI bridge	End-to-end automated pipeline: scanner → AI → client database
Month 3	Implement CER monitoring dashboard on the Data Warehouse	Weekly accuracy reports; automatic flagging of low-confidence document types

Phase 3 — Long-Term (Months 4–12): Scale

Timeline	Action	Strategic Outcome
Month 4	Expand dataset to 15,000+ samples; retrain Florence-2-large for 10 epochs	Model accuracy at production grade for all major document types
Month 6	Evaluate 'DISI Box' product: pre-configured server for client on-premises deployment	New revenue stream: sell managed AI hardware to clients; recurring support contract
Month 6	Apply fine-tuning methodology to Zambia, Zimbabwe, and Botswana document types	DISI's AI capability scales to all six operating countries
Month 9	Migrate Data Lake to a structured Data Warehouse (DuckDB / PostgreSQL) for analytics	SLA performance tracking; accuracy trend reports for management and clients
Month 12	Evaluate the second GPU workstation for redundancy / parallel processing	Zero downtime for production clients; 2× throughput capacity

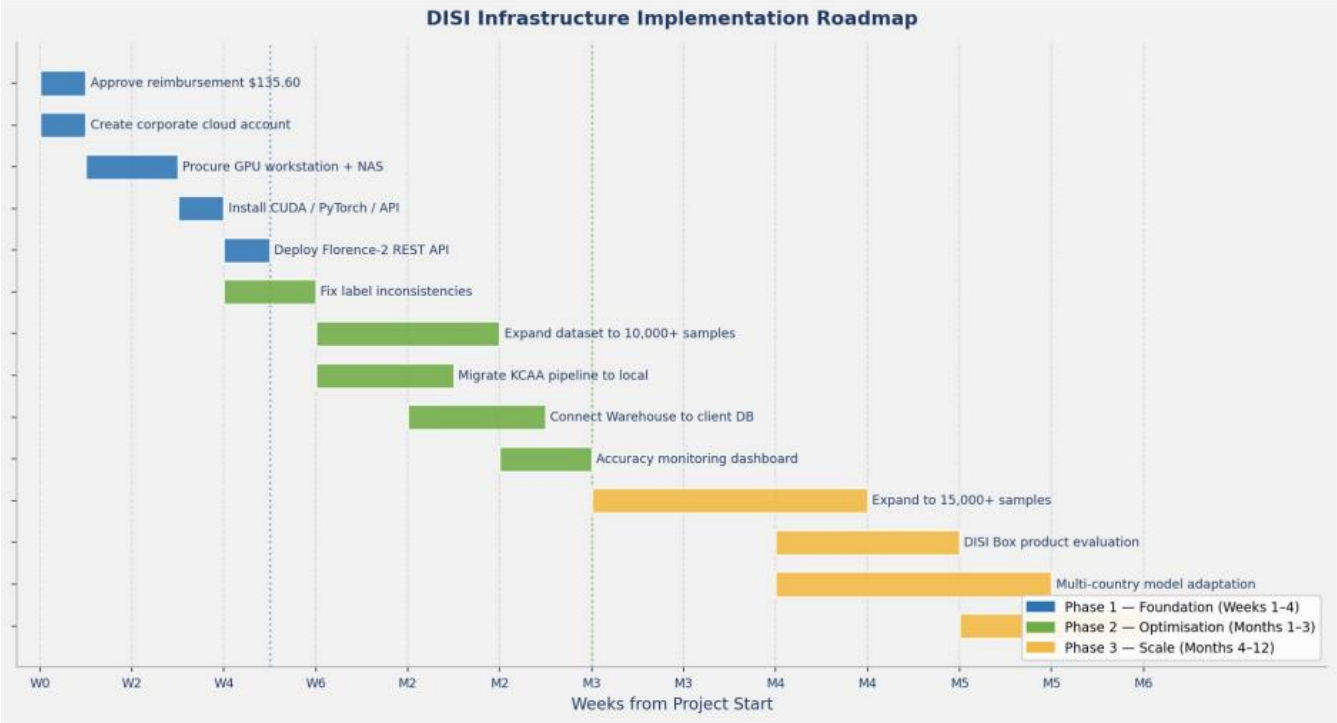

Figure 5: Implementation roadmap. Phase 1 establishes the local infrastructure foundation. Phase 2 optimises accuracy and automates pipelines. Phase 3 scales across countries and document types

13. Staff Training and Knowledge Management

13.1 Roles Required

Role	Responsibility	Training Required
AI Developer (existing)	Model training, fine-tuning, API development, pipeline maintenance	Python, PyTorch, FastAPI, Label Studio — already in place
IT / System Administrator	GPU workstation maintenance, NAS backup, network isolation, and access control	1-week GPU server administration training
Document Operations Staff	Label Studio annotation for model retraining (spot-check semi-automated labels)	2-day Label Studio annotation training
Data / Business Analyst	Warehouse query, accuracy dashboards, client reporting	SQL / MySQL queries; 1-week data reporting training
Project Manager	Client onboarding, SLA monitoring, and escalation of low-accuracy document types	1-day pipeline overview; ongoing dashboard access

13.2 Knowledge Documentation Requirements

- **Model Registry:** All fine-tuned model versions must be stored in the Data Lake with training configuration, evaluation metrics, training data version, and deployment date.

- **Pipeline Documentation:** Every ingest script, FastAPI endpoint, and database schema must be documented in a shared internal wiki (Confluence, Notion, or equivalent).
- **Runbook:** A step-by-step operations runbook must be maintained for GPU workstation startup, model loading, batch job submission, and error recovery.
- **Training Data Versioning:** Every update to the training dataset (new labels, corrected annotations) must be versioned and tagged in the Data Lake with the date and author.

14. Expenditure Summary and Reimbursement Request

14.1 R&D Expenditure Breakdown

Platform	Sessions	Spend	% of Total	Purpose
RunPod GPU Compute	11 sessions	$111.00	84.7%	Model training, benchmarking, evaluation, fine-tuning
Google Colab Pro	1-month subscription	$9.99	7.6%	Early-stage experiments, dataset preprocessing
Google Colab Compute Units	100 units	$9.99	7.6%	Model testing and dataset preparation
Transaction Fees	13 transactions × $0.36	$4.62	3.5%	Non-productive overhead per top-up
GRAND TOTAL	13 sessions	$135.60	100%	All paid personally — reimbursement requested

All 11 RunPod invoices and 2 Google Colab receipts are attached to the original GPU Utilisation Report (submitted separately). The total of $135.60 was paid from personal funds and is requested for reimbursement as legitimate R&D expenditure on behalf of DISI.

14.2 Proposed Infrastructure Investment Summary

Item	Category	Estimated Cost (USD)	Priority
RTX 5090 GPU Workstation (Option A)	CAPEX — Hardware	$2,500 – $3,500	Critical
8 TB NAS / Server Storage (Option A)	CAPEX — Hardware	$1,000 – $2,000	Critical
R&D Reimbursement (personal spend)	Reimbursement	$135.60	Immediate
Corporate Cloud Account (RunPod/Lambda)	OPEX — Monthly	~$50 – $150 / month (burst only)	High
Label Studio + MinIO setup (in-house)	OPEX — One-time	$0 (open-source)	High
Annual Maintenance Budget (hardware)	OPEX — Annual	~$250 / year	Medium
TOTAL YEAR 1 INVESTMENT	—	~$4,000 – $5,800	—

TOTAL YEAR 2+ (annual running cost)	—	~$900 – $2,050 / year	—

15. Conclusion

For a total personal investment of **$130.98**, the NSSF OCR R&D project has proven that a permanently-owned, task-specific AI model can process documents at **6,127 documents per hour** — 5.1× above the production target — for **$0.00024 per document**. That is a 99%+ cost reduction compared to manual data entry, and orders of magnitude cheaper than commercial OCR APIs. The model handles Kenyan abbreviations, degraded scans, mixed handwritten and printed text, and outputs structured JSON directly into SQL and MySQL databases.

This capability currently resides on personal cloud accounts and personal hardware. It is fragile, non-compliant for many clients, and does not scale. Moving it onto **DISI-owned infrastructure** will: eliminate non-productive session overhead; satisfy privacy-sensitive clients who generate the highest-value contracts and cannot currently be served; stop the accumulation of perpetual cloud costs; build a self-improving data asset; and position DISI as the only document management firm in the region that can offer a **fully air-gapped, on-premises AI processing solution**.

DISI has digitised over **600 million documents** across Africa. The proposed investment of $4,000–$5,800 in Year 1 will unlock contracts worth tens or hundreds of thousands of dollars per year — contracts that cloud-dependent competitors simply cannot win.

RECOMMENDED IMMEDIATE ACTIONS:

1. Approve reimbursement of $135.60 in documented personal R&D expenditure.
2. Authorise procurement of one RTX 5090 workstation (~$3,000) and one NAS (~$1,500).
3. Create a corporate DISI cloud account and migrate all client work off personal accounts.
4. Schedule a 1-week infrastructure setup sprint with the AI Developer.

These four actions cost less than one month of equivalent cloud storage alone —
and deliver a permanently-owned AI infrastructure asset.

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